

Experiment No. 1

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August 2026

1 Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data and make predictions or decisions without being explicitly programmed for every task.

Machine Learning is widely used in many fields such as health-care, finance, education, transportation, and entertainment.

2 Types of Machine Learning

There are mainly three types of Machine Learning.

2.1 Supervised Learning

In supervised learning, the machine is trained using labelled data. The algorithm learns the relationship between input and output and uses it to make predictions for new data.

Examples of supervised learning include:

- Linear Regression
- Logistic Regression
- Decision Trees
- Support Vector Machines

2.2 Unsupervised Learning

In unsupervised learning, the machine works with unlabelled data. The algorithm tries to find hidden patterns or groups in the data.

Examples include:

- Clustering
- K-Means Algorithm
- Principal Component Analysis

2.3 Reinforcement Learning

Reinforcement learning is a type of Machine Learning in which an agent learns by interacting with an environment.

The agent receives rewards for correct actions and penalties for incorrect actions.

3 Applications of Machine Learning

Machine Learning has many real-world applications.

3.1 Healthcare

Machine Learning can help doctors analyse medical data, identify diseases, and predict patient outcomes.

3.2 Finance

Banks and financial institutions use Machine Learning for fraud detection, risk analysis, and credit scoring.

3.3 Education

Machine Learning can be used to create personalised learning systems and analyse student performance.

3.4 Entertainment

Streaming platforms use Machine Learning to recommend movies, music, and other content based on user preferences.

4 Advantages of Machine Learning

Some important advantages of Machine Learning are:

- It can analyse large amounts of data quickly.
- It can automate repetitive tasks.
- It can identify complex patterns in data.
- It can improve predictions over time.

5 Conclusion

Thus we have successfully created a document using section, sub-section and text formatting.

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PRN NO-26070521228

18-08-2026